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COX–12

12" Compact Coaxial Line Array Speaker

USER MANUAL

450W RMS Power	**98 dB** Sensitivity	**127 dB** Max SPL Cont.	**100°×10°** Dispersion

For Touring & Fixed Installations

1. Product Overview

The COX–12 is a precision-engineered coaxial line array speaker system purpose-built for high SPL main PA applications in both touring and fixed installations, particularly within small-format environments. Combining professional-grade acoustic performance with a robust and portable design, the COX–12 represents the pinnacle of compact line array engineering.

Equipped with a robust 12-inch (304 mm) coaxial driver and a 1.5-inch (38 mm) high-frequency compression driver, the COX–12 delivers exceptional acoustic performance across the full audible spectrum. The passive, two-way full-range system incorporates a carefully tuned crossover network, making it well-suited for professional sound reinforcement across a range of venues — including nightclubs, auditoriums, multipurpose halls, rental services, and other premium installations.

Product Images

Left: COX-12 Array Stack | Right: Front & Rear Panel Views

2. Key Features

- ▶️ For Touring & Fixed Installations — versatile deployment in any professional environment
- ▶️ True 450W RMS Power (900W Program / 1800W Peak) in passive mode
- ▶️ Crossover Modes: Passive / Bi-Amp Switchable for flexible amplifier configurations
- ▶️ Up to 16 Boxes per line can be installed with full safety compliance
- ▶️ 12" Coaxial driver with 1.5" HF compression driver for true point-source performance

▶️ Integrated rigging system with stainless steel push-pins for fast, secure deployment
▶️ Touring-grade textured black paint finish for durability in demanding environments
▶️ 100°(H) × 10°(V) horn design providing wide horizontal and tight vertical dispersion
▶️ High-density multi-layer plywood enclosure optimized for rental and permanent install
▶️ 98 dB sensitivity with up to 130 dB Peak SPL capability

3. Technical Specifications

Parameter	Specification
Type	Two-way Coaxial full range Bi-amp/Passive switchable line array speaker
Frequency Response (-10 dB)	60 Hz – 19 kHz
LF Drivers	1× 12" (304 mm) Coaxial Driver
HF Drivers	1× 1.5" (38 mm) HF Compression Driver
Power — Passive Mode (RMS)	450W RMS
Power — Passive Mode (Program)	900W
Power — Passive Mode (Peak)	1800W
Power — Bi-amp LF (RMS / Program / Peak)	400W / 800W / 1600W
Power — Bi-amp HF (RMS / Program / Peak)	50W / 100W / 200W
Sensitivity (1W/1m)	98 dB
Max SPL Continuous (1m)	127 dB
Max SPL Peak (1m)	130 dB
Nominal Impedance	8 Ohms
Crossover Mode	Passive / Bi-amp Switchable
Crossover Point — LF (Bi-amp)	65 Hz – 3 kHz
Crossover Point — HF (Bi-amp)	3 kHz – 19 kHz
Dispersion Pattern	100°(H) × 10°(V)
Cabinet Material	High density multi-layer plywood
Dimensions (H × W × D)	350 × 500 × 450 mm
Weight	22 kg / 48.5 lbs

4. Physical Dimensions

Cabinet Dimensions (H × W × D): 350 mm × 500 mm × 450 mm
Weight: 22 kg (48.5 lbs)
Cabinet Material: High-density multi-layer plywood
Finish: Touring-grade textured black paint
Rigging: Integrated stainless steel push-pin locking hardware

The enclosure is constructed from high-density, multi-layer plywood, optimized for both permanent installation and rental use. The cabinet design prioritizes vocal intelligibility, sonic fidelity, and low distortion, while maintaining a responsive transient profile.

5. Crossover & Amplification

5.1 Passive Mode

In Passive Mode, the internal crossover network handles signal division between the LF and HF drivers. A single amplifier channel drives the full system at 450W RMS (8 Ohms). This is the simplest configuration, suitable for standard installations.

5.2 Bi-Amp Mode

In Bi-Amp Mode, the crossover is bypassed. Two separate amplifier channels are required — one dedicated to the LF driver and one to the HF driver. This allows for precise gain control, greater dynamic headroom, and optimized driver protection.

Channel	Frequency Range
LF Channel (Bi-amp)	65 Hz – 3 kHz
HF Channel (Bi-amp)	3 kHz – 19 kHz

⚠️ **Important:**
The Louis Martin digital controller must be programmed by an authorized technician. Do not use Louis loudspeaker systems without the correct controller and the program. Warranty is void if system failure is caused by incorrect controller use.

6. Installation & Rigging

6.1 Rigging System

The COX–12 features an integrated rigging system with stainless steel locking push-pins and adjustable angling hardware. This system allows quick, secure coupling of multiple enclosures into a line array configuration with variable splay angles between adjacent cabinets.

6.2 Array Configuration

- ▶️ Up to 16 boxes can be configured per line with full safety compliance
- ▶️ Push-pin locking ensures enclosures remain securely coupled during flight
- ▶️ Adjustable angling hardware allows customization of array curvature to suit venue geometry
- ▶️ Always use certified flying hardware rated to a minimum of 10× the total system weight

6.3 Ground Stacking

The COX–12 can be ground-stacked in pairs or small arrays for smaller events. Ensure the surface is level and load-bearing. Use anti-slip material under the bottom enclosure when stacking.

7. Connection & Wiring

7.1 Connector Panel

The rear panel provides input connectivity for both Passive and Bi-Amp operating modes. The Passive / Bi-Amp switch must be set correctly before powering the amplifier.

7.2 Passive Mode Wiring

- ▶️ Connect the amplifier output to the input terminal labeled FULL RANGE
- ▶️ Ensure the Passive/Bi-Amp switch is set to PASSIVE
- ▶️ Use professional-grade speaker cable of appropriate gauge for run length

7.3 Bi-Amp Mode Wiring

- ▶️ Set the Passive/Bi-Amp switch to BI-AMP before making connections
- ▶️ Connect amplifier channel 1 (LF) to the LF input terminal
- ▶️ Connect amplifier channel 2 (HF) to the HF input terminal

- ▶️ Set crossover points on the amplifier or DSP controller: LF at 65 Hz–3 kHz, HF at 3 kHz–19 kHz
- ▶️ Use the Louis Martin digital controller for optimized system tuning

8. Safety Instructions

READ ALL SAFETY INSTRUCTIONS BEFORE USE
Failure to follow these instructions may result in personal injury, equipment damage, or voided warranty.

8.1 Flying & Suspension Safety

- ▶️ When flying speakers, appropriate certified hardware and professional rigging techniques must be employed for safely suspending the enclosures.
- ▶️ All speaker hardware and flying apparatus must be regularly inspected for any visual or mechanical failure before each use.
- ▶️ Do not exceed the maximum array configuration of 16 boxes per line.
- ▶️ All rigging must be performed by a qualified rigging professional.
- ▶️ Maintain adequate clearance below the array for audience safety.

8.2 Audience & Venue Safety

- ▶️ Leave adequate distance between speakers and the audience. Refer to your local authority for Health and Safety guidance when using loudspeaker systems.
- ▶️ Ensure all cable runs are properly secured and do not create tripping hazards.
- ▶️ Excessive SPL levels can cause permanent hearing damage. Follow local noise ordinance regulations.

8.3 Electrical Safety

- ▶️ Protect your speakers and electronics from exposure to humidity, water, or rain. The COX–12 is not rated for outdoor use without appropriate weatherproofing.
- ▶️ Ensure amplifiers are matched to the speaker's impedance and power ratings.
- ▶️ Always power down amplifiers before making or changing speaker connections.

8.4 Controller Safety

- ▶️ The Louis Martin digital controller should only be programmed by an authorized technician.
- ▶️ Do not use Louis Martin loudspeaker systems without the correct controller and approved program loaded.
- ▶️ Any system failure due to incorrect controller use will void the product warranty.

9. Maintenance & Care

9.1 Routine Inspection

▶️ Inspect all connectors, cables, and rigging hardware before each deployment
▶️ Check all push-pin locking mechanisms for wear or deformation
▶️ Inspect the cabinet for any cracks, delamination, or structural damage
▶️ Verify grille integrity and clean with a soft damp cloth when necessary

9.2 Storage

▶️ Store in a dry, climate-controlled environment away from direct sunlight
▶️ Use padded flight cases or protective covers for transportation
▶️ Avoid stacking more than 4 enclosures high during storage
▶️ Protect connectors with dust caps when not in use

9.3 Driver Care

▶️ Do not operate the system at power levels beyond rated specifications
▶️ Allow adequate thermal recovery time after extended high-SPL operation
▶️ Contact an authorized Louis Martin service center for driver replacement

10. Warranty Information

Warranty Period: 2 Years from Original Date of Purchase
Coverage: Defects in workmanship or materials
Exclusions: User abuse, accidents, unauthorized modifications, incorrect controller use
Service: Defective items will be repaired or replaced at no charge for materials or labour
Process: Items must be adequately packed and sent pre-paid to an authorized Louis Martin distributor or service center

The loudspeakers and electronics are covered against defects in workmanship or materials for a period of two years from the original date of purchase. The warranty does not cover any item which in Louis Martin's opinion has failed due to user abuse, accident, or modifications.

11. Technical Support & Contact

Contact	Details
Website	www.louismartinaudio.it
Email	contact@louismartinaudio.it
Technical Support	contact@louismartinaudio.it
Warranty Claims	Contact your authorized Louis Martin distributor

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